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Understanding the effects of long-term different fertilizer applications on methylmercury accumulation in rice (*Oryza sativa* L.) plants

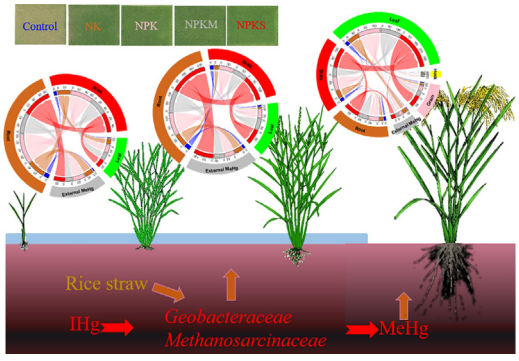
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HIGHLIGHTS

- 24 years fertilizer and organic amendments in rice paddy system were studied.
- *Geobacteraceae* and *Methanosarcinaceae* were the main Hg methylating microorganism in this study.
- Chemical fertilizer with rice straw return (NPKS) promoted the growth of *Geobacteraceae and Methanosarcinaceae*.
- NPKS led to the highest MeHg concentration in rice grain and paddy soil.

GRAPHICAL ABSTRACT



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ABSTRACT

The effects of long-term applications of fertilizer form and rate on methylmercury concentration in rice grain have been previously reported. However, there remains a large gap in understanding fertilizer treatments affect the transformation of mercury (Hg) in paddy soils and the translocation of Methylmercury (MeHg) in rice plants. Thus, we investigated the effects of four fertilizer treatments (NK, nitrogen and potassium; NPK, nitrogen, phosphorus and potassium; NPKM, nitrogen, phosphorus, potassium and swine manure; and NPKS, nitrogen, phosphorus, potassium and rice straw) on mercury (Hg) methylation in paddy soils and on MeHg translocation in rice plants after a 24-year period; specifically, we investigated differences in MeHg production in paddy soils and MeHg mass flow in rice plants. The results showed that *Geobacteracea* and *Methanosarcinacea* were the major Hg-methylating microorganisms and their relative abundances were significantly promoted by rice straw return, which led to the significantly higher MeHg concentration both in paddy soils and in the rice grain of NPKS treatment compared with control. In addition, the phosphate and organic fertilizers increased MeHg translocation from the roots to the leaves and grain more than the nitrogen and potassium fertilizers did. In detail, compared with the control, NK increased MeHg concentrations in the rice grain by approximately 1-fold; NPK and NPKM, by approximately 3-fold; and NPKS, by approximately 8-fold. Last, we suggest that policy-makers worldwide should comprehensively assess the high potential risk of MeHg in rice grain in large production areas where rice straw is returned.

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1. Introduction

Given that rice (*Oryza sativa* L.) and rice-derived food are the staple food for more than 50% of the global population (Rothenberg and Feng, 2012), increased accumulations of methylmercury (MeHg) in rice, which has high neurotoxicity and bioaccumulation effects (Cheng et al., 2006), have attracted worldwide attention (Qiu et al., 2008; Qiu et al., 2006). To date, the consumption of rice has been an important pathway by which humans in Hg mining areas in China are exposed to MeHg (Feng et al., 2008; Li et al., 2008; Meng et al., 2014; Meng et al., 2010; Meng et al., 2011). It is well documented that MeHg in rice grain is mainly taken up by the roots (Qiu et al., 2008; Zhang et al., 2010), and flooding rice paddies strongly promotes the growth of anaerobic microbes that can induce the transformation of inorganic mercury (IHg) to MeHg (Liu et al., 2014; Parks et al., 2013). Therefore, the diversity of anaerobic microbes within and physicochemical properties of paddy soil play a very important role in promoting MeHg accumulation in rice plants (Wang et al., 2014a; Wang et al., 2014b; Zhang et al., 2014).

Notably, long-term applications of fertilizer can alter soil microbe structure, the pH and other soil physicochemical properties (Barkay et al., 1997; Miskimmin et al., 1992), thus likely affecting the bioavailability and potential methylation of Hg. In addition, the use of different fertilizers also results in large variations in rice growth conditions and grain yield (Deng et al., 2010; Liao et al., 2013), possibly leading to different patterns of MeHg translocation in rice plants. Our previous study (Tang et al., 2018) indicated that different long-term fertilizers induce large variations in MeHg rice grain; specifically, high amounts of MeHg present in the rice grain were induced by rice straw return. However, it still has a large gap in understanding the fertilizer treatments working on MeHg formation in paddy soils or MeHg uptake, translocation and accumulation during different growth periods of rice plants. Therefore, we investigated a 24-year-long fertilizer experiment involving four fertilizer treatments in a rice paddy field reported by our previous study (Tang et al., 2018) to test the mechanism of MeHg accumulation in different fertilizer treatments. Specifically, we studied the differences in Hg-methylating microorganisms and MeHg mass flow in rice plants and ultimately discussed the potential risk of different fertilizer applications.

2. Materials and methods

2.1. Site descriptions and experimental design

This study was carried out at a monitoring station of long-term fertilizer applications in Chongqing, China (30°26′ N, 106°26′ E), which is a part of the Long-Term National Soil Fertilizer Monitoring Network (LTNSFMN). This long-term fertilizer experiment has been ongoing since 1991 including 9 fertilizer group treatments. Based on previous study (Tang et al., 2018), four fertilizer treatments and a control treatment (unfertilized) were chosen to demonstrate the effects of chemical fertilizers (urea, calcium superphosphate and potassium sulfate) and organic amendments (swine manure and rice straw) on soil fertility and environmental influences. The four treatments included the following: (1) urea and potassium sulfate (NK); (2) urea, calcium superphosphate and potassium sulfate (NPK); (3) urea, calcium superphosphate, potassium sulfate and swine manure (NPKM); and (4) urea, calcium superphosphate, potassium sulfate and rice stalk (NPKS). The description for the planting pattern and fertilizer applications was shown in detail in our earlier study (Tang et al., 2018). Briefly, Chemical fertilizers were applied twice a year before planting wheat and rice, respectively. Swine manure and rice straw were applied once a year before planting wheat. During 1991–1996, application of chemical fertilizer was N 150 kg ha^{-1}, P 32.8 kg ha^{-1}, K 62.2 kg ha^{-1}; since fall of 1996, application of chemical fertilizer was changed to N 135 kg ha^{-1}, P 26.2 kg ha^{-1}, K 49.8 kg ha^{-1}. Organic amendments were remaining unchanged with swine manure 22.5 × 103 kg ha^{-1} y^{-1} and rice straw (dry weight) 7.5 × 103 kg ha^{-1} y^{-1}.

2.2. Sample collection

The rice was grown in nursery from April 12 to May 10, 2015. Rice plant and soil samples to determine Hg species and soil physicochemical properties were collected on May 10, June 10, July 10 and August 10, 2015, which represented the rice seedling, tillering, heading and ripening stages, respectively. A detailed description of the sample collection and preprocessing is available in our previous study (Tang et al., 2018). Briefly, our studied paddy field was 120 m^2 and was divided into three subplots for replication. For sampling, we collected samples from with the 0–5-cm, 5–10-cm, 10–20-cm and 20–30-cm soil profiles. The rice plants and corresponding soil samples were sealed in polypropylene bags and then rapidly transported to the lab for further treatment. The soil samples were freeze-dried and ultimately passed through a 100-mesh sieve to determine the total mercury (THg) and MeHg. Soil particles were washed off the rice plant samples with deionized water. Afterward, the samples were freeze-dried; divided into rice root, stalk, leaf, husk and grain subsamples; and ultimately ground to pass through a 100-mesh sieve to determine THg and MeHg.

Soil samples (0–10 cm depth) for determining soil microorganisms were collected at the rice heading stage (July 10, 2016). The soil samples were divided into two batches in an anaerobic operating box. One batch was used for DNA extraction. The other was air dried to determine the moisture coefficient and then passed a 60-mesh sieve for analyses of the cation exchange capacity (CEC), sodium hydroxide-hydrolyzable nitrogen (HN), sodium bicarbonate-extractable phosphorus (EP), sodium bicarbonate-extractable potassium (EK), diethylenetriaminepentaacetic acid (DTPA), extractable iron (EFe) and extractable manganese (EMn). A small portion of the air-dried sample was further ground and passed through a 100-mesh sieve for analyses of total nitrogen (TN), total phosphorus (TP) and organic carbon (OC). The determining method was showed in Text S1 of supplementary information (SI).

2.3. DNA extraction, MiSeq sequencing, and bioinformatics analysis

Soil DNA was extracted from 0.3–0.4 g of mixed fresh soil with a Fast DNA SPIN Kit (MP Biomedicals, Illkirch, France) in an anaerobic operation box. The DNA quality was evaluated by a Nanodrop 2000 (DU 800, Beckman Coulter, USA) with 260/280 and 260/230 ratios and then stored at −80 °C for further analysis. The bacterial and archaeal 16S rRNA genes were amplified using 515F/907R (5′-TGCCAGCMGCCGCGG-3′/5′-CCGTCAATTCMTTTRAGTTT-3′) and Arch519F/Arch915R (5′-CAGCCGCCGCGGTAA-3′/5′-GTGCTCCCCCGCCAATTCCT-3′) primers, respectively. PCR was performed according to the methods of a previous study (Xiang et al., 2018). The prepared samples were detected with MiSeq by Majorbio BioTech Co., Ltd.

Raw sequence data were obtained using the Mothur (v.1.39.5, https://www.mothur.org/wiki/Download_mothur) program (Kozich et al., 2013). Sequences with >97% identity and <3% divergence were grouped into the same Operational Taxonomic Unit (OTU). The obtained sequences were compared with the sequence of the V4–V5 region (start = 11,894, end = 27,654) of the Silva v.132 reference file database (https://www.mothur.org/wiki/Silva_reference_files) and the Ribosomal Database Project (RDP) v.16 (Trainset 16_022016.rdp, https://www.mothur.org/wiki/RDP_reference_files) database to obtain taxonomic information (Cole et al., 2013).

2.4. Quality control

Quality control measures consisted of method blanks, triplicates, matrix spikes, and certified reference materials (Table S1) as described in the Text S2 of SI.

2.5. Statistical analysis

Venn diagrams were constructed in R with the ggplot2 package to demonstrate the common and unique OTUs of soil bacteria and archaea in response to the different fertilizer applications. Redundancy analysis (RDA) between Hg-methylating microorganism and MeHg, or selected soil factors were conducted in R with the vegan package to reveal relative importance of Hg-methylating microorganism that govern Hg-methylating, or relative importance of soil factors that govern Hg-methylating microorganism in soil. A MeHg mass flow diagram was constructed in R with the circlize package to show the translocation of MeHg in the rice tissue, and detailed information can be found in Text S3, Table S1, and S2 of the Supplementary Information (SI). A structural equation model (SEM) was constructed with Amos 24 to demonstrate the effects of rice tissue biomass on MeHg concentration. Mean differences among treatments were determined using one-way ANOVA with Duncan's test at $p < 0.05$ in SPSS 22.0.

3. Results

3.1. Soil physicochemical property changes after long-term fertilizer applications

The soil physicochemical properties were different after 24-year fertilizer applications (Table 1). The average of soil total nitrogen (TN), total phosphorus (TP), total potassium (TK) and organic carbon (OC) in paddy soil ranged from 1.08 g kg^{-1} to 1.59 g kg^{-1}, 0.45 g kg^{-1} to 0.96 g kg^{-1}, 17.8 g kg^{-1} to 19.0 g kg^{-1}, and 12.9 g kg^{-1} to16.0 g kg^{-1}, respectively (Table 1). All fertilizer treatments have led to a significant increase of TN concentration ($p < 0.05$, by one-way ANOVA) compared with the initial TN concentration in 1991 (1.25 ± 0.11 g N kg^{-1}) or Control (1.08 ± 0.03 g N kg^{-1}), but a decrease of TK compared with the initiate of this experiment in 1991 (21.1 ± 0.38 g K kg^{-1}), especially, the lowest TK concentration found in NPKS (17.8 ± 0.6 g K kg^{-1}). Soil TP in treatments with calcium superphosphate (NPK, NPK, and NPKS) was significantly higher than the initiating TP concentration in 1991 (0.67 ± 0.06 g P kg^{-1}) and of treatments without calcium superphosphate (NK: 0.45 ± 0.02 g P kg^{-1} and Control: 0.55 ± 0.22 g P kg^{-1}). The soil pH value in Control (7.7) was observed without distinct variation, but in the rest fertilizer treatments (including NK, NPK, NPKM, and NPKS) was significantly decreased from 7.5 to 6.8. The soil cation exchangeable capacity (CEC) was observed without significant differences among each fertilizer treatment and Control.

The average of soil hydrolyzed nitrogen (HN), extractable phosphorus (EP), extractable potassium (EK), extractable iron (EFe) and extractable manganese (EMn) ranged from 83.2 mg N kg^{-1} to 121.5 mg N kg^{-1}, 0.6 mg P kg^{-1} to 42.5 mg P kg^{-1}, 58.3 mg K kg^{-1} to 78.3 mg K kg^{-1}, 37.5 mg Fe kg^{-1} to135.5 mg Fe kg^{-1}, and 7.6 mg Mn kg^{-1} to18.9 mg Mn kg^{-1}, respectively (Table 1). The HN concentrations in treatments with nitrogen fertilizers (NK, NPK, NPKM, and NPKS) were significantly higher ($p < 0.05$, by one-way ANOVA) than treatments without chemical fertilizers (Control). Same trend was observed in phosphorus, i.e., the EP concentrations in NPK, NPKM and NPKS were significantly higher ($p < 0.05$, by one-way ANOVA) than in NK and Control.

3.2. Hg species variations in the soil

The average soil MeHg concentration during the rice growth period was in the range of 0.20 to 4.71 μg kg^{-1} (Fig. 1a, b, c, and d). For all treatments, the MeHg concentration gradually decreased from the 0–5 cm surface soil to the 20–30 cm deep soil, suggesting net MeHg formation occurred mainly in the upper soil layer, especially in 0–10 cm. The soil MeHg concentrations in all the treatments generally increased from the rice seedling stage to the ripening stage. MeHg concentrations in control and NPKS were significantly lower than those in the NPK, and NPKM treated 0–30 cm soil, and NK treated 5–30 cm soil at the rice seedling stage ($p < 0.05$; Fig. 1a, b, c, d). From the rice tillering stage, the MeHg concentrations in NK, NPK and NPKM were comparable to those in the NPKS in 0–5 cm soil (Fig. 1a), while the MeHg concentration in NPKS was significantly greater than that in the control in 0–20 cm soil ($p < 0.05$, Fig. 1a, b, c). Specifically, MeHg concentrations in NPKS promptly increased and were greater than those in the other treatments beginning at the rice heading stage (Fig. 1a, b, c and d).

THg concentrations in the 0–20-cm soil layer was greater in the fertilizer treatments than in the control treatment, except in the NK-treated 0–20-cm soil layer in the ripening stage (Fig. 1e, f, and g). In addition, we found that THg in the paddy soil profile gradually migrated into the paddy water, as evidenced by on one hand, an increase of THg and MeHg concentration in overlying water, interface water and pore water from rice seedling stage to heading stage (SI: Fig. S3); on the other hand, a decrease in the soil profile THg concentration from the rice seedling stage to the heading stage and an increase in the THg concentration during the rice ripening stage when the paddy field water dried out. Specifically, the most significant increase in the soil profile THg concentration in NPKS during the ripening stage suggested that the rice straw contributed the greatest dissolved Hg in the rice paddy water.

3.3. Bacterial and archaeal community structures after long-term fertilizer applications

Information about bacterial and archaeal community structures after long-term fertilizer applications is shown in Figs. S1 and S2. Eight hundred and fifty-one operational taxonomic units (OTUs) of bacteria,

Table 1
The soil physicochemical properties under different fertilizers*.

Treatments†	TN	TP	TK	OC	pH	CEC	HN	EP	EK	EFe	EMn
	g N kg^{-1}	g P kg^{-1}	g K kg^{-1}	mg C kg^{-1}		cmol kg^{-1}	mg N kg^{-1}	mg P kg^{-1}	mg K kg^{-1}	mg Fe kg^{-1}	mg Mn kg^{-1}
Control	1.08 ± 0.03^{d}	0.55 ± 0.22^{b}	17.9 ± 1.8^{a}	13.9 ± 1.6^{ab}	7.7 ± 0.1^{a}	104.3 ± 20.9^{a}	83.2 ± 3.6^{b}	0.9 ± 0.7^{c}	63.0 ± 5.5^{b}	37.5 ± 4.8^{b}	9.5 ± 0.3^{b}
NK	1.33 ± 0.02^{c}	0.45 ± 0.02^{b}	19.0 ± 0.9^{a}	13.9 ± 2.7^{ab}	6.9 ± 0.6^{b}	120.5 ± 9.1^{a}	112.6 ± 4.5^{a}	0.6 ± 0.2^{c}	78.3 ± 4.3^{a}	135.5 ± 16.9^{a}	18.9 ± 1.8^{a}
NPK	1.31 ± 0.09^{c}	0.77 ± 0.03^{a}	18.5 ± 0.8^{a}	12.9 ± 0.8^{b}	7.5 ± 0.4^{ab}	117.9 ± 4.1^{a}	108.6 ± 7.6^{a}	28.8 ± 1.2^{b}	58.3 ± 8.3^{b}	110.0 ± 18.5^{a}	13.0 ± 4.2^{b}
NPKM	1.43 ± 0.04^{b}	0.96 ± 0.03^{a}	18.2 ± 0.1^{a}	14.9 ± 1.1^{ab}	6.8 ± 0.4^{b}	133.5 ± 1.3^{a}	121.5 ± 11.7^{a}	42.5 ± 2.2^{a}	68.5 ± 6.9^{b}	124.2 ± 15.1^{a}	13.3 ± 4.0^{b}
NPKS	1.59 ± 0.06^{a}	0.89 ± 0.36^{a}	17.8 ± 0.6^{a}	16.0 ± 0.4^{a}	7.0 ± 0.4^{b}	133.5 ± 6.2^{a}	121.4 ± 8.4^{a}	41.3 ± 2.6^{a}	64.0 ± 11.5^{b}	110.4 ± 11.2^{a}	7.6 ± 2.3^{b}

TN: total nitrogen; TP: total phosphorus; TK: total potassium; OC: organic carbon; pH:soil:water = 1:2.5; CEC: Cation Exchange Capacity; HN: NaOH hydrolysable nitrogen; EP: $NaHCO_3$ extractable phosphorus; EK: ammonium acetate extractable potassium; EFe: diethylene-triaminepentaacetic acid (DTPA) extractable iron; and EMn: DTPA extractable manganese.

* Lower case letters indicate significant difference at 95% significant level according to Duncan test.

† Control: no fertilizer or organic amendments; NK: urea and potassium sulfate; NPK: urea, calcium superphosphate and potassium sulfate; M: swine manure only; and S: rice stalk.

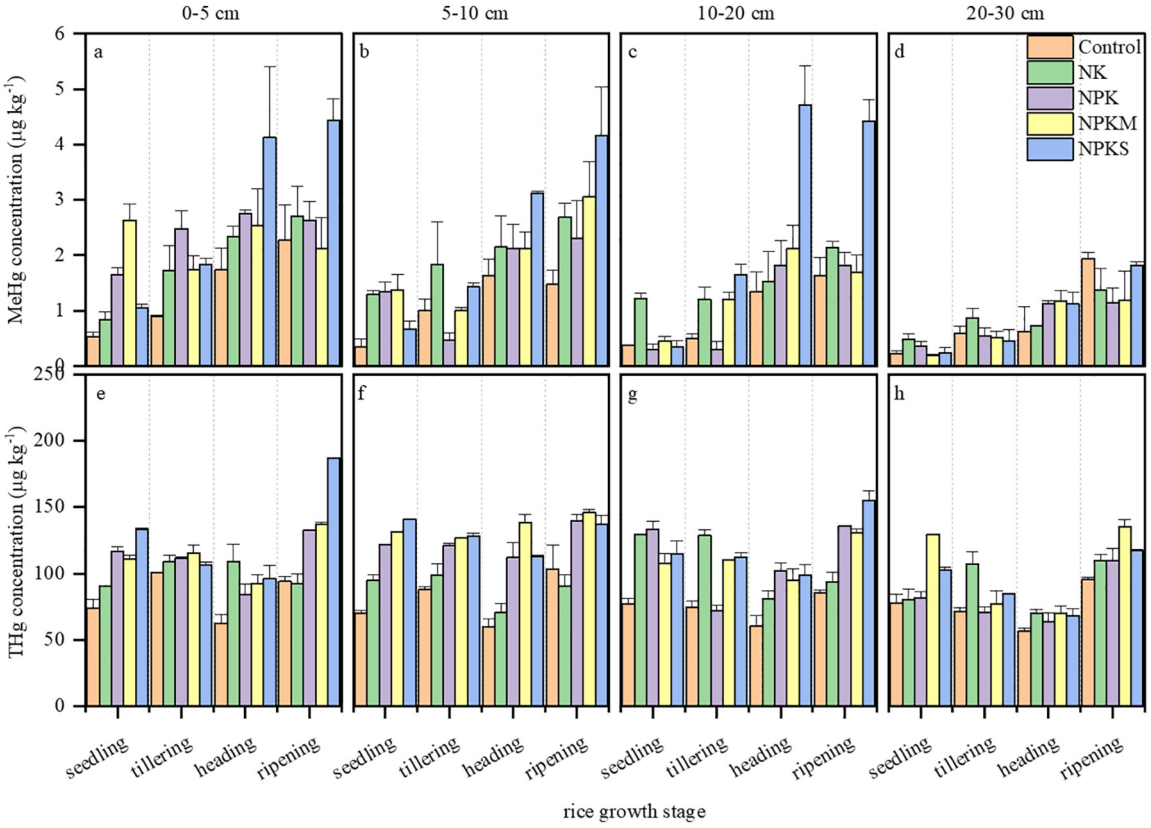


Fig. 1. Concentrations of MeHg and THg within the 0–30-cm soil profile in different fertilizer-treated paddy soils during the rice growing period. Figures "a" and "e" show the variations in MeHg and THg in the 0–5 cm soil layer, respectively; "b" and "f", variations in MeHg and THg in the 5–10 cm soil layer, respectively; "c" and "g", variations in MeHg and THg in the 10–20 cm soil layer, respectively; and "d", and "h", variations in MeHg and THg in the 20–30 cm soil layer, respectively. Different letters above the bar indicate significant differences of soil Hg concentration among treatments in same rice growth period at $p < 0.05$ according to Duncan test. Treatments are described in the footnote of Table 1.

and 37 OTUs of archaea were detected in four-fertilizer and Control treated soils during rice heading stage. Those bacteria could be classified into 32 phyla (Fig. S2a). The highest relative abundance of bacteria in all treatments was *Proteobacteria*, followed by *Acidobacteria* and *Chloroflexi*, which accounting for 31.6%–35.7%, 12.0%–14.0%, and 10.2%–10.8% of total bacterial OTUs numbers, respectively. In addition, the *Actinobacteria*, *Firmicutes*, *Bacteroidetes*, *Planctomycetes*, *Gemmatimonadetes* and *Nitrospirae* also had relative abundances of bacteria more than 1%. The relative abundances of the remainder 23 phyla were less than 1%. The unclassified bacteria in each treatment accounted for 19.5%–20.4% of the total bacterial community. The classified archaea were distributed in 7 phyla (Fig. S2b). The highest relative abundance was *Thaumarchaeota*, accounting for 50.3%–75.0% of the total archaeal OTUs numbers. In addition, the other dominant archaea included *Woesearchaeota*, *Crenarchaeota*, *Euryarchaeota*, and *Pacearchaeota*, which taking 2.8%–5.0%, 3.7%–14.0%, 4.4%–8.5%, and 3.3%–5.9% of total archaea, respectively. The relative abundances of *Diapherotrites* and *Aenigmarchaeota* were less than 0.2%. The unclassified archaea accounted for 7.5%–16.3% of the total archaeal community.

The largest difference in the bacterial and archaeal community structures occurred between NPKS and the control. The NPKS treatment presented the greatest relative abundance of *Proteobacteria*, *Bacteroidetes*, *Crenarchaeota* and *Euryarchaeota*, while the control presented the greatest relative abundance of Acidobacteria and Firmicutes. From Fig. S1 of SI, there were 459 shared bacterial OTUs out of a total of 851 bacterial OTUs, and most of them were in the *Proteobacteria* (208 OTUs) and *Firmicutes* (54 OTUs) phyla. Additionally, 25, 43, 22, 44 and 32 unique bacterial OTUs were detected in the control, NK, NPK, NPKM and NPKS treatments, respectively (SI: Fig. S1). The unique bacteria in NPKS and the control were in the Proteobacteria phylum; in NK and NPK, the Proteobacteria, Firmicutes and Actinobacteria phyla; and in NPKM, the Firmicutes phylum. For archaea, twenty-nine shared archaeal OTUs of a total of 37 archaeal OTUs were detected, and most belonged to the *Methanomicrobia* (SI: Fig. S1). Most of the treatments lacked unique archaea, except for NPK and NPKM, each of which contained one unique archaeon (*Methanolobus* and *Methanomethylovorans*, respectively).

3.4. Hg uptake and translocation in rice plants

The average MeHg concentration in all the rice roots decreased by 73.6%–94.1% from 5.73 μg kg^{-1} during the seedling stage to 0.34–1.51 μg kg^{-1} during the ripening stage (Fig. 2a, b, c, d, and e). MeHg concentrations in the fertilizer-treated rice roots (NK, NPK, NPKM and NPKS) were significantly greater than those in the control rice roots during the rice tillering stage to the ripening stage. We also observed decreasing trends for MeHg concentrations in the rice stalks, with the exception of NPK, in which the MeHg concentration presented a slight increase beginning at the rice seedling stage. In contrast to that in the rice roots, the average MeHg concentrations in the rice stalks in the NPKM and NPKS treatments were lower than those in the other fertilizer treatments and the control treatment during the rice tillering stage to the heading stage. The average MeHg concentration in the rice leaves increased by 96.6%–545.5% from 0.88 μg kg^{-1} during the seedling stage to 1.73–5.68 μg kg^{-1} during the heading stage, but then sharply decreased by 91.2%–96.2% to 0.15–0.25 μg kg^{-1} during the ripening stage. The average MeHg concentrations in the leaves of phosphorus-treated rice plants (NPK, NPKM, and NPKS) were significantly greater than those

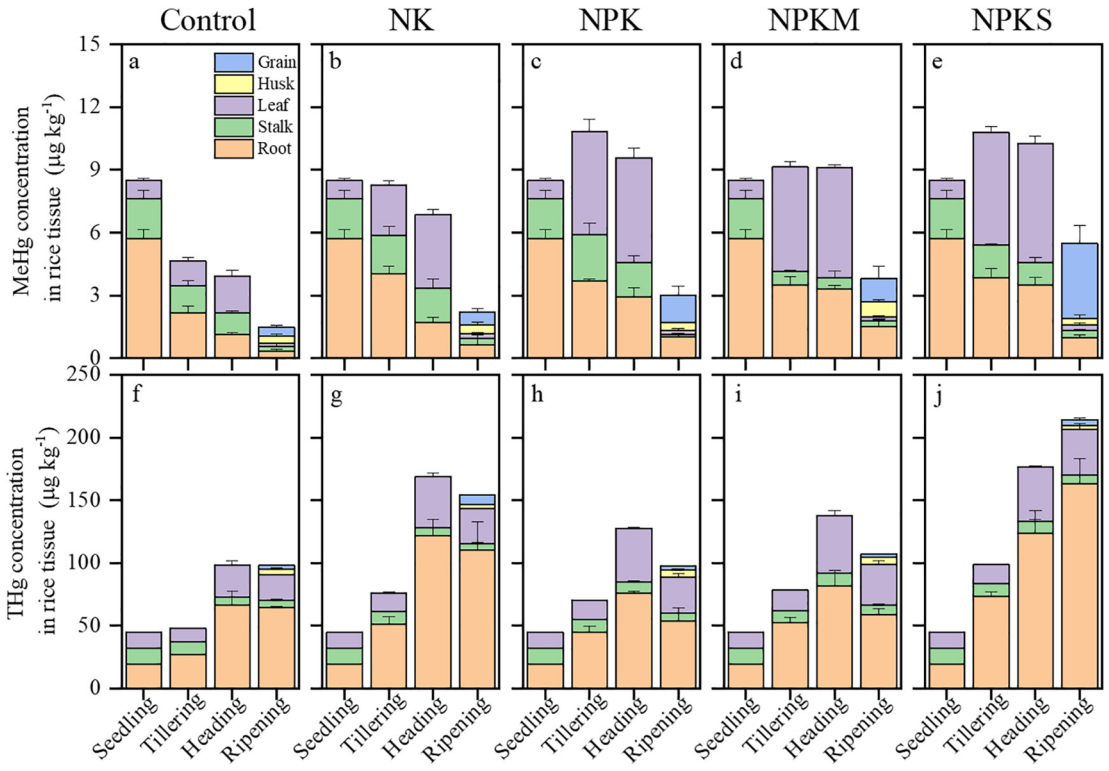


Fig. 2. Concentrations of MeHg and THg in tissue samples of rice plants treated with different fertilizers during the growing period. Treatments are described in the footnote of Table 1.

without phosphorus-treated rice plants (NK and control) during the rice tillering stage to the heading stage. For the husks, we found only that NPKM significantly promoted the MeHg concentration, which was 2-fold greater than that in the control. The MeHg concentration in rice grain was reported in our previous study (Tang et al., 2018). Briefly, the average MeHg concentration in the rice grain was in the range of 0.43–3.58 μg kg^{-1}. Compared with the control treatment, all the fertilizer treatments presented significantly greater MeHg concentrations in the rice grain ($p < 0.05$, one-way ANOVA). Specifically, the average MeHg concentration in the grain of NPKS-treated rice was significantly higher than that of the remainder treatments ($p < 0.05$, one-way ANOVA).

In contrast to MeHg in rice plants being derived mainly from the soil, IHg allocation in rice is controlled by both root uptake of Hg(II) from the soil and leaf uptake of Hg^0 from the atmosphere. Our previous study (Tang et al., 2018) discussed this issue in detail, but the present study focused mainly on MeHg formation and transportation in rice paddies. Hence, the description of THg variation under long-term fertilizer use (Fig. 2f, g, h, i, and j) is discussed in detail in text S4 of the SI. Briefly, during the rice tillering stage, the average THg concentrations in the roots of rice plants in the fertilizer treatments (NK, NPK, NPKM and NPKS) were significantly greater than those in the control treatment ($p < 0.05$, one-way ANOVA). In addition, we found that all fertilizers significantly increased the THg concentration in rice leaves ($p < 0.05$, one-way ANOVA), while NK significantly increased the THg concentration only in the rice grain.

4. Discussion

4.1. Dominant Hg-methylating microorganisms

The MeHg concentration in paddy soils is governed mainly by Hg-methylating microorganisms (Hu et al., 2013; Lu et al., 2016), including sulfate-reducing bacteria (SRB), iron-reducing bacteria (FeRB), methanogens and other microorganisms (SI: Table S4) (Börjesson et al., 2012; Figueiredo et al., 2015; Gilmour et al., 2013; Lu et al., 2016; Miskimmin et al., 1992; Parks et al., 2013; Podar et al., 2015; Yu et al., 2013; Zhong et al., 2010). Of these Hg-methylating microorganisms, SRB are most well documented; for example, treatment with sulfate-reducing bacterium inhibitors can significantly decrease the production of MeHg in paddy soils (Tang et al., 2019). In the present study, we detected not only a sulfate-reducing bacterial community (*Syntrophobacteraceae*, *Syntrophacaeae*, *Desulfuromonadaceae*, *Desulfovibrionaceae*, and *Desulfovibrionaceae-unclassified*; Fig. 3), but also other Hg-methylating microorganisms (*Geobacteraceae*, *Ruminococcaceae*, *Bacillaceae*, *Veillonellaceae*, *Mycobacteriaceae*, *Methanospirillaceae*, and *Methanosarcinaceae*; Fig. 3). The relative abundances of *Geobacteraceae* (0.91%–2.56%), *Bacillaceae* (0.65%–1.54%) and *Methanosarcinaceae* (0.55%–1.9%) were significantly greater than the relative abundance of SRB (0.33%–1.38%) (Fig. 3, $p < 0.05$).

Redundancy analysis showed that the soil MeHg concentrations were positively related with to *Geobacteracea*, *Ruminococcaceae*, *Syntrophobacteraceae*, *Methanospirillaceae*, and *Methanosarcinaceae*, and negatively related to *Bacillaceae*, *Desulfuromonadaceae*, *Veillonellaceae*, and *Mycobacteriaceae* (Fig. 4a). However, of those Hg-methylation microorganism, only *Geobacteracea* ($r = 0.750$, $p < 0.01$) and *Methanosarcinaceae* ($r = 0.658$, $p < 0.01$) were significantly positively correlated with soil MeHg concentration. Additionally, *Xanthomonadales* and *Chitinophagaceae*, which were reported in predicting MeHg production as the non-methylated microorganism (Liu et al., 2019), was also detected in this study. Unfortunately, no significantly correlation was observed between MeHg concentration and the relative abundances of *Xanthomonadales* ($r = 0.132$, $p > 0.05$) or *chitinophagaceae* ($r = -0.163$, $p > 0.05$). Therefore, unlike previous studies highlighting the role of SRB, *Xanthomonadales* or *Chitinophagaceae*, our study suggests that *Geobacteraceae* and *Methanosarcinaceae* are the main Hg-methylating microorganisms at our study site. This conclusion is consistent with the results of a study by Liu et al. (2016).

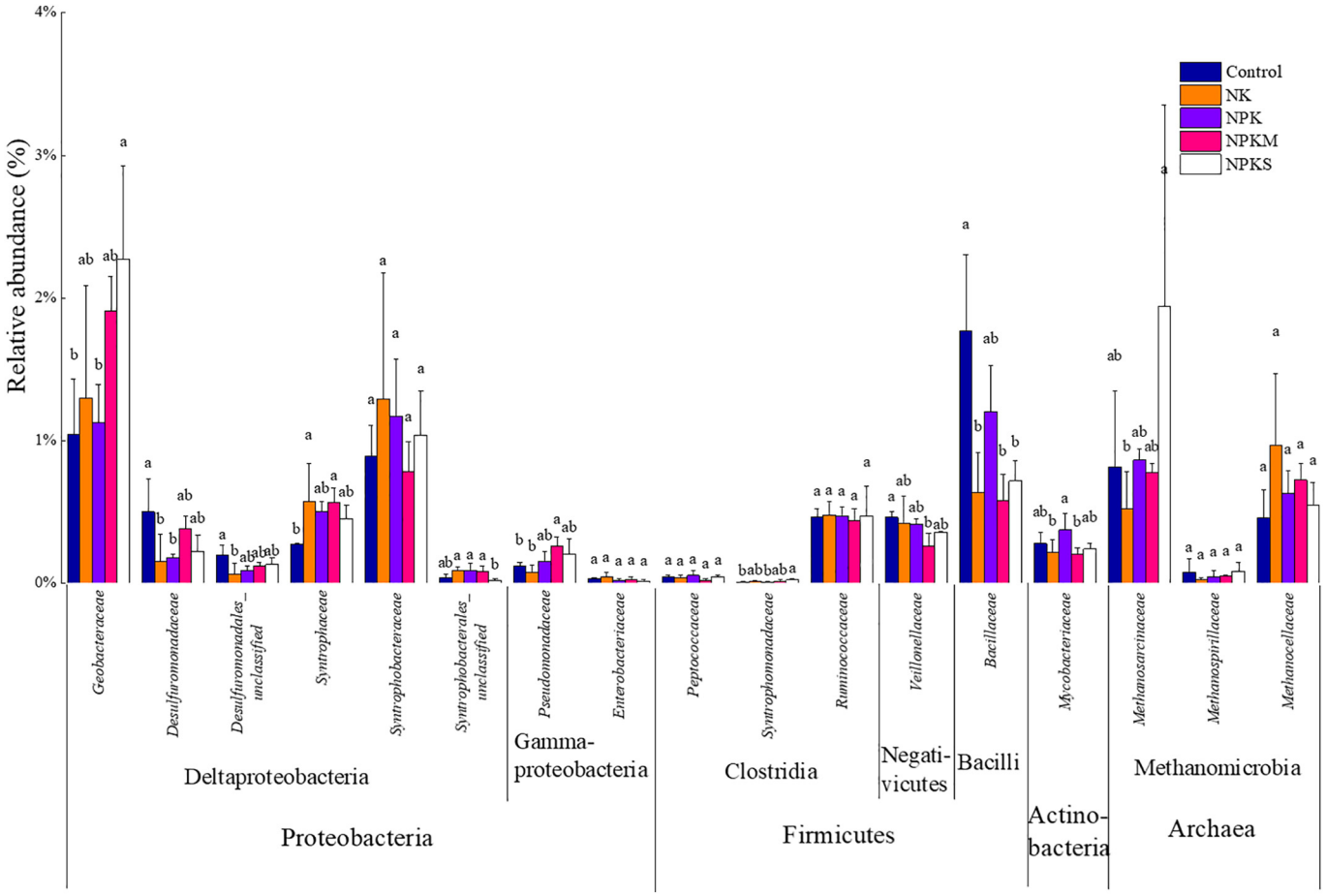


Fig. 3. Relative abundance of bacteria and archaea with the ability to methylate Hg in response to different fertilizers. Treatments are described in the footnote of Table 1. Different letters above the bar indicate significant differences of relative abundance of bacteria or archaea among treatments at $p < 0.05$ according to Duncan test.

In the present study, two genera of bacteria (*Geobacter* and *Geobacteraceae_unclassified*) and four genera of archaea (*Methanosarcina*, *Methanolobus*, *Methanomethylovorans*, and *Methanosarcinaceae_unclassified*) were detected in communities of *Geobacteracea* and *Methanosarcinaceae*, respectively. *Geobacter*, which accounted for 98% of the *Geobacteracea*, is a typical genus of FeRB, nearly all the species of which (except for *Geobacter lovleyi*) contain homologous genes of *hgc*AB and thus can efficiently convert

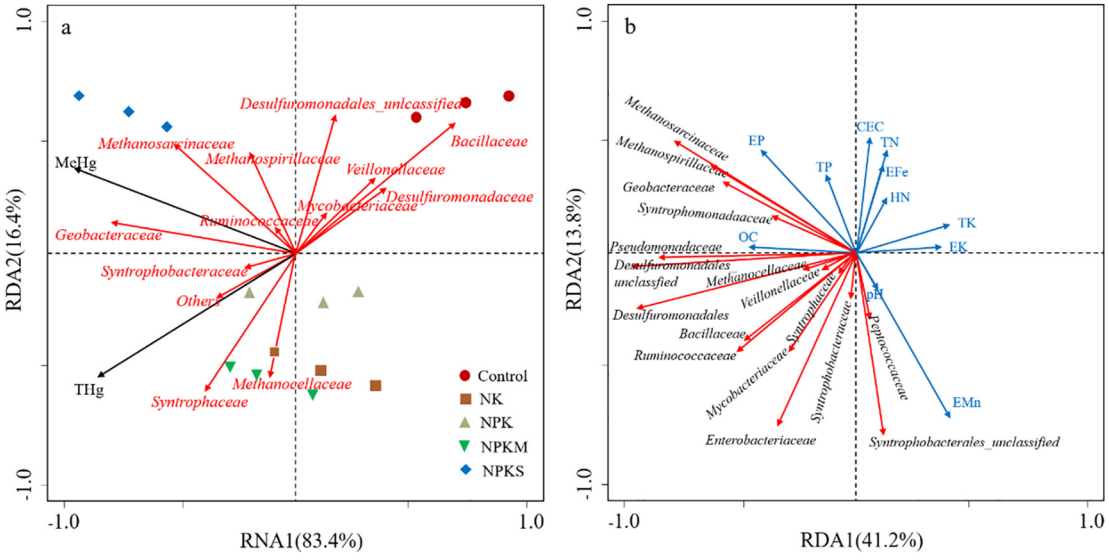


Fig. 4. (a) Relationships among Hg-methylating bacteria and archaea, MeHg concentration and THg concentration; (b) relationships between bacterial and archaeal community structures and physicochemical properties in soils treated with different long-term fertilizer applications.

inorganic or elemental Hg to MeHg (Fleming et al., 2006; Gilmour et al., 2013; Hu et al., 2013; Ma et al., 2019; Schaefer and Morel, 2009). With respect to the *Methanosarcinaceae* community, *Methanolobus* cannot transform inorganic Hg to MeHg (Gilmour et al., 2013) but can produce acetate, which can be a primary energy source for some microorganisms, such as FeRB (of which *Geobacter* species are members) and SRB (Kerdchoechuen, 2005; Liu and Whitman, 2008). *Methanomethylovorans* (Hu et al., 2013) and *Methanospirillum* (Yu et al., 2013) have been suggested to possess the *hgc*AB gene. Hence, the presence of *Geobacteraceae* and *Methanosarcinaceae* communities can significantly improve MeHg production in paddy soils.

4.2. Effects of different long-term fertilizer applications on Hg methylation in paddy soils

Compared with the NK treatment, the NPK treatment led to the significant increase of THg concentration in the 0–5-cm soil layer ($p < 0.05$), but no significant difference were observed MeHg concentration between NK treatment and NPK treatment (Fig. 1). These findings are consistent with those of our previous studies showing that the calcium superphosphate fertilizer does not significantly increase the content of bioavailable Hg (Tang et al., 2018). Compared with NPK, the addition of swine manure (NPKM) also did not increase the 0–5-cm soil THg and MeHg concentrations. The rice straw return treatment (NPKS) led to a significant increase of the 0–5 cm soil THg and MeHg concentrations than control and other fertilizer treatments. This result can be explained by two causes. One is that the higher THg concentration in paddy water of NPKS treatment than that in the other treatments (Fig. S3) indicated that long-term rice straw return could increase the concentration of dissolved Hg in the soil, thus increasing bioavailable Hg and promoting soil MeHg production (Tang et al., 2018). The other is that NPKS possibly induces a greater relative abundance of Hg-methylating microorganisms. In addition, we found that long-term fertilizer applications significantly altered the soil physicochemical properties. Hence, we hypothesized that long-term fertilizer-induced changes in soil physicochemical properties can significantly affect differences in Hg-methylating microorganisms and MeHg concentrations in paddy soils. This hypothesis can be supported by the results of the redundancy analysis shown in Fig. 4a and b.

First, compared with the other treatments, the control treatment promoted the growth of Hg-methylating microorganisms such as *Bacillaceae*, *Desulfuromonadaceae*, *Mycobacteriaceae*, *Veillonellaceae*, and *Desulfuromonadales_unclassified* but limited the growth of *Geobacteracea* and *Syntrophaceae* (Figs. 4a, and 3).

Second, the addition of nitrogen, potassium and phosphorus fertilizer potentially promoted the growth of bacteria for nitrate reduction (Zhang et al., 2018) and *Geobacteraceae* organisms, which would compete with various SRB for labile carbon (Zhang et al., 2018). Thus, we found that soil HN and EK were significantly negatively correlated with *Desulfuromonadales_unclassified*, *Syntrophaceae* and *Desulfuromonadaceae* (Fig. 4b; Table S5), but significant positive correlations between *Geobacteraceae* and both EP and TN were detected (Fig. 4b; Table S5). This is the reason for the decreased relative abundances of *Desulfuromonadaceae* and *Desulfuromonadales_unclassified* and the increased relative abundance of *Geobacteraceae* in NK and NPK compared with the control (Fig. 3). In addition, *Syntrophobacterales_unclassified* was significantly correlated with EMn (Table S5, $p < 0.01$), and the nitrogen, potassium and phosphorus fertilizers that induced a distinct increase in EMn (Table 1) also promoted the growth of *Syntrophobacterales_unclassified* in NK and NPK. The absolute values of most correlation coefficients (0.5 to 0.6) suggest that nitrogen, potassium and phosphorus fertilizers cannot completely change the bacterial and archaeal structures to those dominated by a greater portion of Hg-methylating microorganisms. Fig. 3 shows that the relative abundances of *Geobacteraceae* and *Methanosarcinaceae* increased but did not reach a significant level ($p < 0.05$, one-way ANOVA). Therefore, MeHg production within the 0–5-cm soil layer in the NK and NPK treatments were comparable to that in the control treatment.

Last, *Geobacteraceae*, *Methanospirillaceae*, *Methanosarcinaceae* and *Syntrophomonadaceae* were significantly and positively correlated with soil OC, EP and TP but negatively correlated with pH (Fig. 4b; Table S5), which is consistent with the results of other studies (Liu et al., 2018; Liu et al., 2016). Compared to swine manure and the other chemical fertilizers, the rice straw resulted in the greatest soil OC (16.0 mg C kg^{-1}), greatest TN, and second greatest TP contents and a relatively low pH (Table 1). Moreover, the rice straw could stimulate the growth of *Methanosarcinaceae* and *Methanospirillaceae* to accelerate the formation of acetic acid and other products, which could further promote the growth of *Geobacteraceae* (Lu et al., 2015). Thus, the greatest relative abundances of *Geobacteraceae*, *Methanosarcinaceae* and *Methanospirillaceae* occurred in NPKS (Figs. 3, and 4a). Given that the *Geobacteraceae* and *Methanosarcinaceae* communities compose the

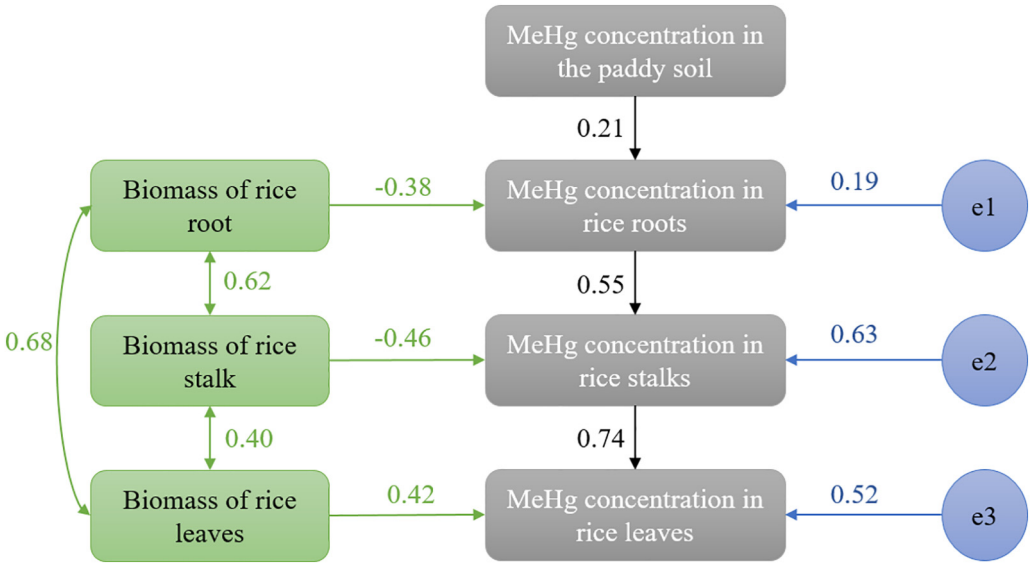


Fig. 5. Contributions of soil MeHg and rice tissue biomass to MeHg concentrations according to structural equation modeling. The SEM was conducted with Amos 24. The numbers near the arrows mean the standardized path coefficient (i.e., the direct effect). The one-way arrows represent the direct effect of one variable on another; the two-way arrows represent the collinearity between the different rice tissue biomass values. The rectangle and ellipses represent the observable variables and residual errors, respectively.

dominant group of Hg-methylating microorganisms, NPKS increased the soil MeHg concentration the most, which reached a significant level in all the fertilizer treatments (Fig. 1). In addition, compared with NPK, swine manure addition (NPKM) did not significantly alter the soil physicochemical properties or the relative abundance of the Hg-methylating microorganisms, and thus the MeHg concentrations in these treatments were comparable.

4.3. MeHg translocation and demethylation in rice plants

Fig. 5 shows the direct effects of root MeHg on stalk MeHg (0.55) and of stalk MeHg on leaf MeHg (0.74). These results suggest that MeHg is distinctly translocated from the roots to leaves during the rice vegetative growth period. During the rice ripening stage, the MeHg stored in the rice leaves was translocated mostly to the rice grain, as evidenced by a sharp decrease in MeHg in the leaves and an increase in MeHg in the rice grain from the heading stage to the ripening stage (Fig. 2; Table S1). In addition, the increased MeHg concentration in the leaves further induced increased MeHg concentrations in the grain ($p < 0.01$, $r = 0.841$; SI: Fig. S5a). Studies have shown that the translocation of MeHg in rice plants is controlled mainly by sulfur-containing amino acids, especially glutathione and cysteine (Meng et al., 2014; Tang et al., 2019; Xu et al., 2016). The translocation of these nutrients in rice plants likely simultaneously drives MeHg mass flow.

The total mass of MeHg in rice plants of each treatment during the rice ripening stage was lower than that during the rice heading stage (external MeHg in SI: Fig. S4c; SI: Table S2); specifically, the mass of MeHg in rice plant were lower 5.3 ± 3.5 ng MeHg plant^{-1} in the control, 48.6 ± 15.7 ng MeHg plant^{-1} in NK, 40.8 ± 7.3 ng MeHg plant^{-1} in NPK, 17.4 ± 11.5 ng MeHg plant^{-1} in NPKM, and 11.9 ± 5.8 ng MeHg plant^{-1} in NPKS. There are two reasons to explain this decrease. One is that the rice residue returns to the paddy surface, leading to MeHg loss. The other is the process of MeHg demethylation in the rice plants (Li et al., 2016; Xu et al., 2016), as several studies have reported distinct MeHg demethylation in rice plants (Strickman and Mitchell, 2017; Xu et al., 2016). Generally, photodegradation (Seller et al., 1996), microbial reductive demethylation (Schaefer et al., 2004) and microbial oxidative demethylation (Marvin-DiPasquale and Oremland, 1998) are considered the three main processes. For rice plants, many studies have focused on *mer*B transgenes to induce the demethylation of MeHg (Arthur et al., 2005; Heaton et al., 2003; Li and Tse, 2015; Wagner-Döbler, 2003). Notably, recent studies have shown that rice plants without transgenes can also demethylate MeHg (Strickman and Mitchell, 2017; Xu et al., 2016). For example, Strickman and Mitchell (2017) speculated that demethylation in rice could be affected by UV light, since MeHg in illuminated straw decreased by approximately 45% compared to 22% in roots without illumination. However, large knowledge gaps concerning the mechanism of demethylation in rice plants exist, and many more studies on this topic are needed.

4.4. Effects of different long-term fertilizers on MeHg allocation in rice plants

First, the effects of long-term fertilizer use on the growth of rice plant organs are considered an important factor in shaping MeHg concentrations in rice plants (Huang et al., 2018; Rothenberg et al., 2012). Fig. 5 shows that the increase in root and stalk biomass can dilute MeHg concentrations directly by factors of 0.38 and 0.46, respectively, while the increase in leaf biomass can increase MeHg concentrations directly by a factor of 0.42 (Fig. 5). This is because many more nutrients are produced and exchanged during the process of photosynthesis in the leaves, possibly resulting in incidental MeHg translocation via sulfur-containing amino acids to the rice leaves.

The different fertilizer treatments can lead to the various generation of glutathione, cysteine and other sulfur-containing compounds in rice plants (Bloem et al., 2010; Carrasco-Gil et al., 2012; Giansoldati et al., 2012; Lundegårdh et al., 2008; Ning et al., 2010; Tripathi et al., 2008), which could affect the translocation of MeHg in rice plant. This phenomenon is supported by long-term NK, NPK, NPKM and NPKS applications causing MeHg concentrations to be greater in the treated rice plants than in the control plants, suggesting that long-term fertilizer use can increase MeHg transport from the soil to rice plants. Specifically, we highlight the important role of organic amendments and phosphorus fertilizer on MeHg accumulation in rice plants. Previous studies have shown that organic amendments can cause rice plants to finish tillering in advance (Belefant-Miller, 2007), and treatments without phosphorus could restrict the growth of rice roots to postpone the growth of rice leaves (Bates and Lynch, 1996; Kirk, 1997). The reason for NK resulting in the allocation of a relatively large amount of MeHg to the roots during the growth of the nutritive organs (before the rice heading stage) is because of the shortage of phosphorus in the rice plants; this phenomenon is in contrast to the much greater MeHg fractions stored in the rice leaves in response to NPKM and NPKS because of the effects of organic amendments and phosphorus fertilizer. For example, during the rapid-leaf growth stage (during the rice tillering stage to the heading stage), the rice plants in NK took up a net 48.88 ng MeHg plant^{-1}, and only 2.9% of the MeHg was allocated to the leaves, while 46.8% of the 59.69 ng MeHg plant^{-1} taken up by the plants in NPKM and 51.9% of the 116.59 ng MeHg plant^{-1} taken up by the plants in NPKS stored the MeHg in their leaves (SI: Table S2; SI: Fig. S3b). The organic amendments and phosphorus fertilizer ultimately lead to a distinct accumulation of MeHg in the grain, as reflected by concentrations that were 3- to 8-fold greater in the treatments than in the control (Fig. 2).

Last, several previous studies involving pot experiments in the lab (Liu et al., 2016; Shu et al., 2016; Zhu et al., 2015) and those based on short-term field surveys (Tang et al., 2019) reported that straw return could induce greater MeHg concentrations in paddy soils and in rice grain. The results of our long-term fertilizer experiment showed that such a phenomenon was particularly evident. NPKS simultaneously had the greatest MeHg in both the paddy soil (Fig. 1) and rice plants (Fig. 2; SI: Table S2) during the rice tillering stage to the heading stage. This is because of the combined effects of the rice straw and chemical fertilizers, as discussed above. More importantly, compared with the control, NK, and NPKM, NPKS increased the MeHg concentration in the grain by a factor of 3, 6 and 8, respectively. Notably, rice straw return played the most important role in increasing MeHg accumulation in the rice grain, followed by phosphorus fertilizer, nitrogen fertilizer and potassium fertilizer. Interestingly, rice straw and swine manure fertilizer have distinctly different effects on leaf MeHg translocation during the rice heading stage to the ripening stage. We found that the addition of swine manure promoted MeHg translocation to the husks, while rice straw did not increase the MeHg concentration in the husks (Fig. 2d and e). This result may be caused by a different pattern of nutrient allocation elicited by swine manure fertilizer, and we recommend additional studies to focus on this issue.

5. Environmental implications

Our results from a 24-year-long cultivation experiment showed that, compared with the control, all the long-term fertilizers induced greater MeHg concentrations in the rice grain. Compared with the other fertilizer or amendments in this study, rice straw treatment played a greater role in increasing the MeHg concentration in both the paddy soil and the rice grain, as well as the relative abundance of *Geobacteracea* and *Methanosarcinaceae*. Notably, straw return in agricultural fields is being encouraged worldwide because of its ability to reduce air pollution from combustion and greenhouse gas (CH_4, and NO_x) emissions (Ma et al., 2009; Watanabe et al., 1993), improving soil OC and soil structure (Devêvre and Horwáth, 2000; Wang et al., 2016; Wang et al., 2015). Based on our finding, the worldwide large-scale straw return to the paddy field, especially in China, could elevate the risk of MeHg exposure. Recent studies showed the biochar, including rice plant-derived

biochar, could mitigate the accumulation of Hg in rice (Shu et al., 2016; Tang et al., 2020; Wang et al., 2019; Xing et al., 2020). Therefore, the future policy formulations and promotions should comprehensively assess the potential MeHg exposure risks caused by rice straw return, and consider reducing rice straw return into the paddy field or converting rice straw into other materials, such as biochar, before returning to the field.

CRediT authorship contribution statement

Zhenya Tang: Conceptualization, Investigation, Writing – original draft. **Fangling Fan:** Investigation, Formal analysis. **Xun Wang:** Conceptualization, Writing – review & editing. **Xiaojun Shi:** Investigation. **Dingyong Wang:** Conceptualization, Writing – review & editing.

Declaration of competing interest

The authors declare no competing interests.

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Appendix A. Supplementary data

Sample determining method, Quality control, translocation of MeHg in rice plant, THg variations during rice growth period, Tables (S1–S5) and Figs. (S1–S5). Supplementary data to this article can be found online at doi: https://doi.org/10.1016/j.scitotenv.2021.146125.

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